



US012387884B2

(12) **United States Patent**
Brower

(10) **Patent No.:** **US 12,387,884 B2**

(45) **Date of Patent:** **Aug. 12, 2025**

(54) **CIRCUIT INTERRUPTING DEVICE HAVING
PRINTED CIRCUIT BOARD COILS**

(58) **Field of Classification Search**

CPC H02H 71/125; H02H 9/50; H05K 1/165
See application file for complete search history.

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(21) Appl. No.: **18/673,963**

(22) Filed: **May 24, 2024**

(65) **Prior Publication Data**

US 2024/0312735 A1 Sep. 19, 2024

Related U.S. Application Data

(63) Continuation of application No. 17/985,758, filed on
Nov. 11, 2022, now Pat. No. 11,996,257, which is a
continuation of application No. 16/520,588, filed on
Jul. 24, 2019, now Pat. No. 11,501,931.

(60) Provisional application No. 62/703,106, filed on Jul.
25, 2018.

(51) **Int. Cl.**

H01H 71/12 (2006.01)

G01R 31/327 (2006.01)

H01H 9/50 (2006.01)

H01H 71/08 (2006.01)

H05K 1/16 (2006.01)

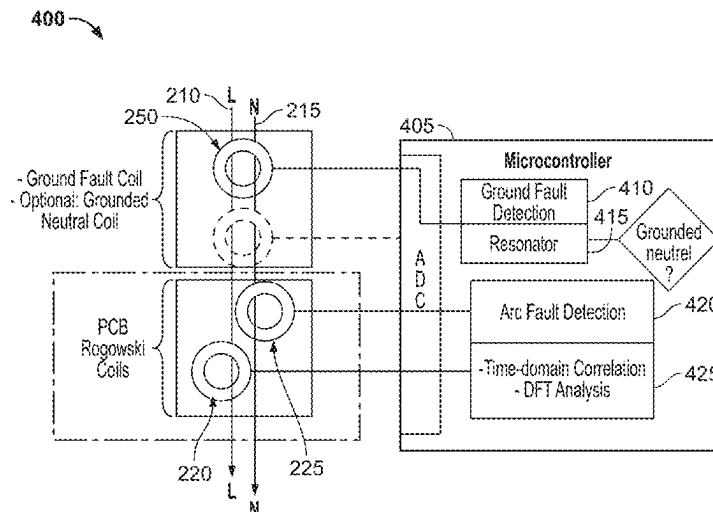
(52) **U.S. Cl.**

CPC **H01H 9/50** (2013.01); **G01R 31/3277**
(2013.01); **H01H 71/08** (2013.01); **H01H**
71/125 (2013.01); **H05K 1/165** (2013.01);
H05K 2201/09027 (2013.01); **H05K**
2201/09063 (2013.01)

(57) **ABSTRACT**

A testing system may include a printed-circuit board coil
embedded on a printed circuit board. The testing system may
further include a test circuit electrically connected to the
printed-circuit board coil, the test circuit configured to:
receive a signal from the printed-circuit board coil, analyze the
signal from the printed-circuit board coil, and determine
a fault based on the signal from the printed circuit board coil.

20 Claims, 8 Drawing Sheets



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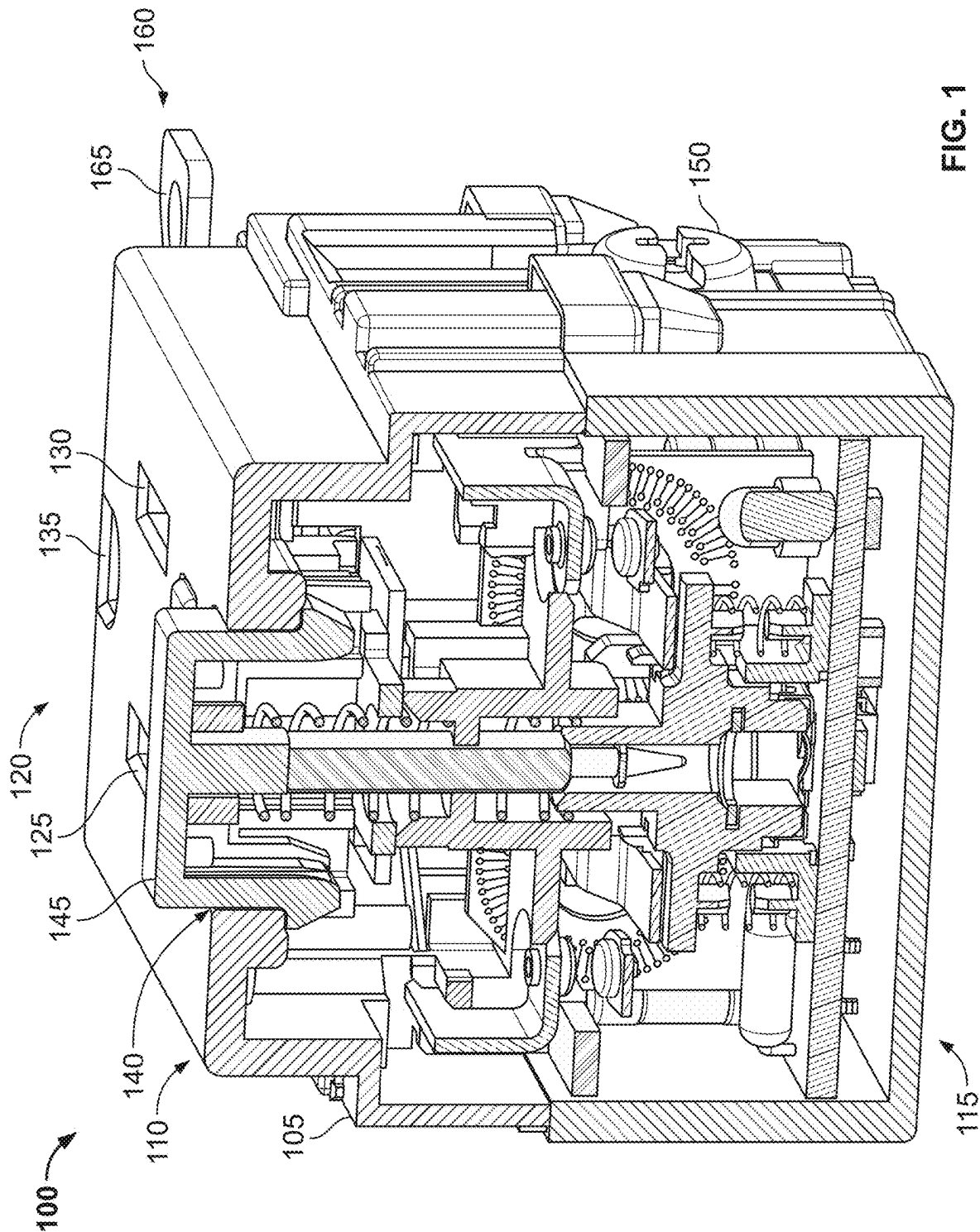
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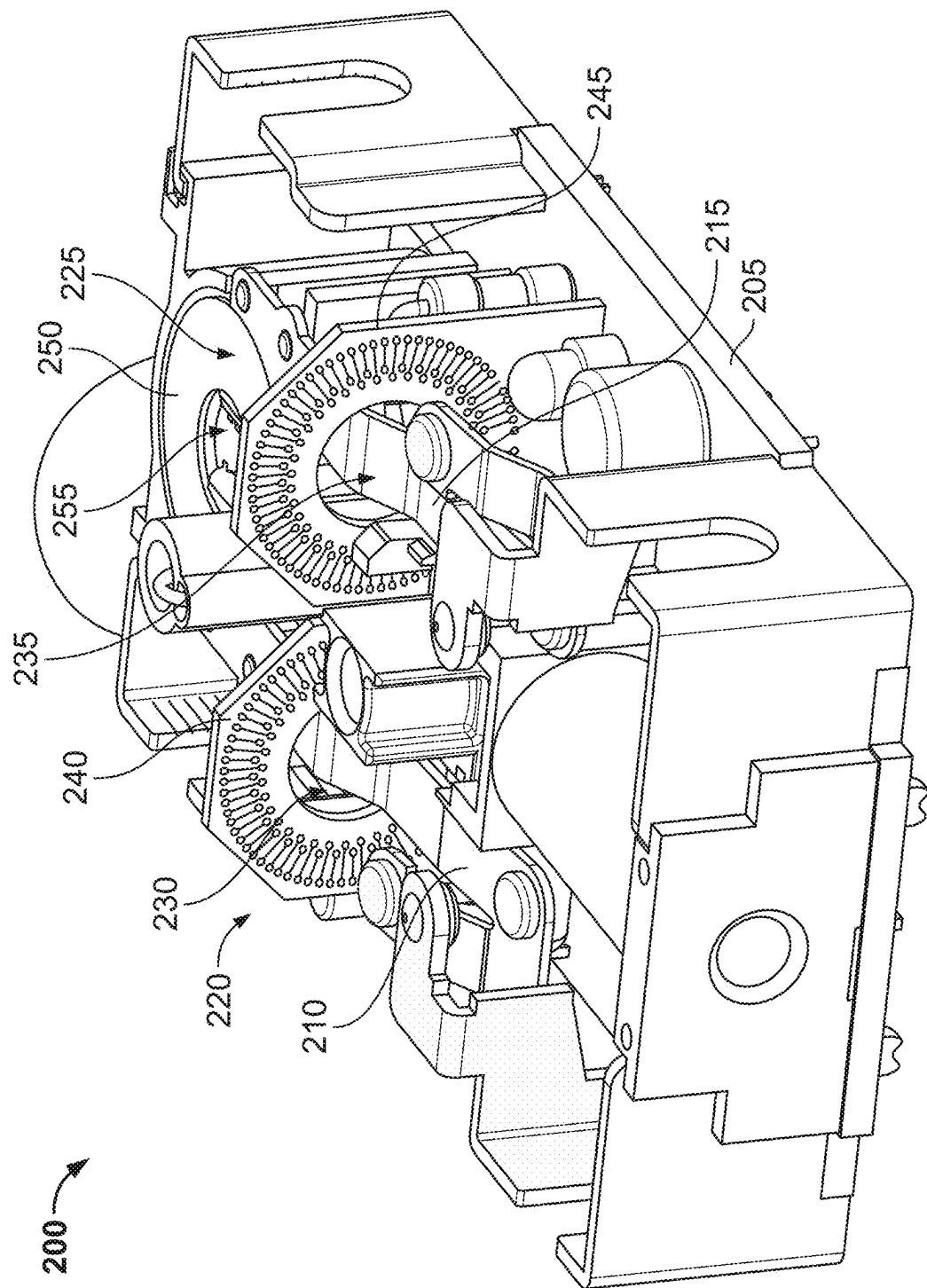


FIG. 2A

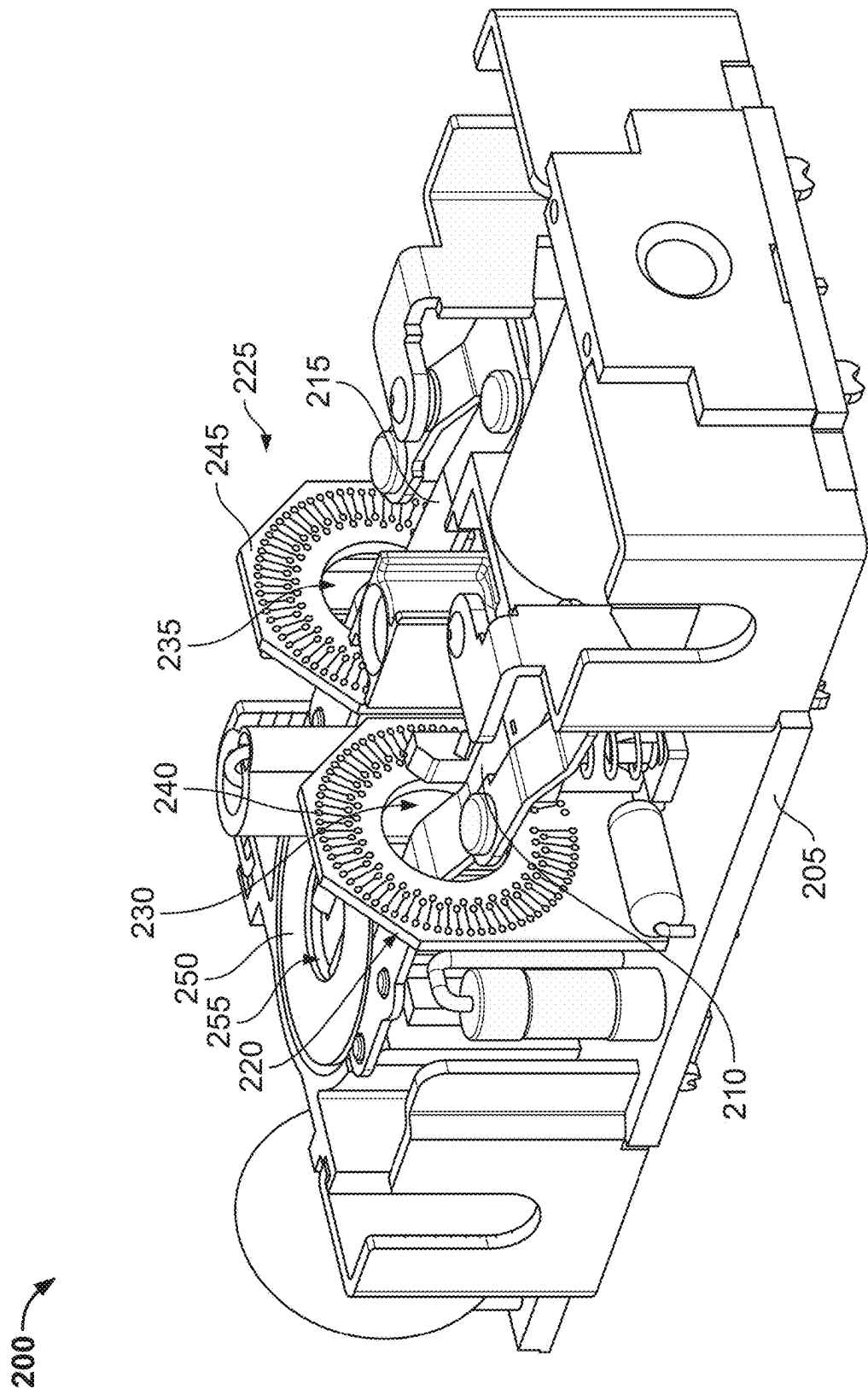


FIG. 2B

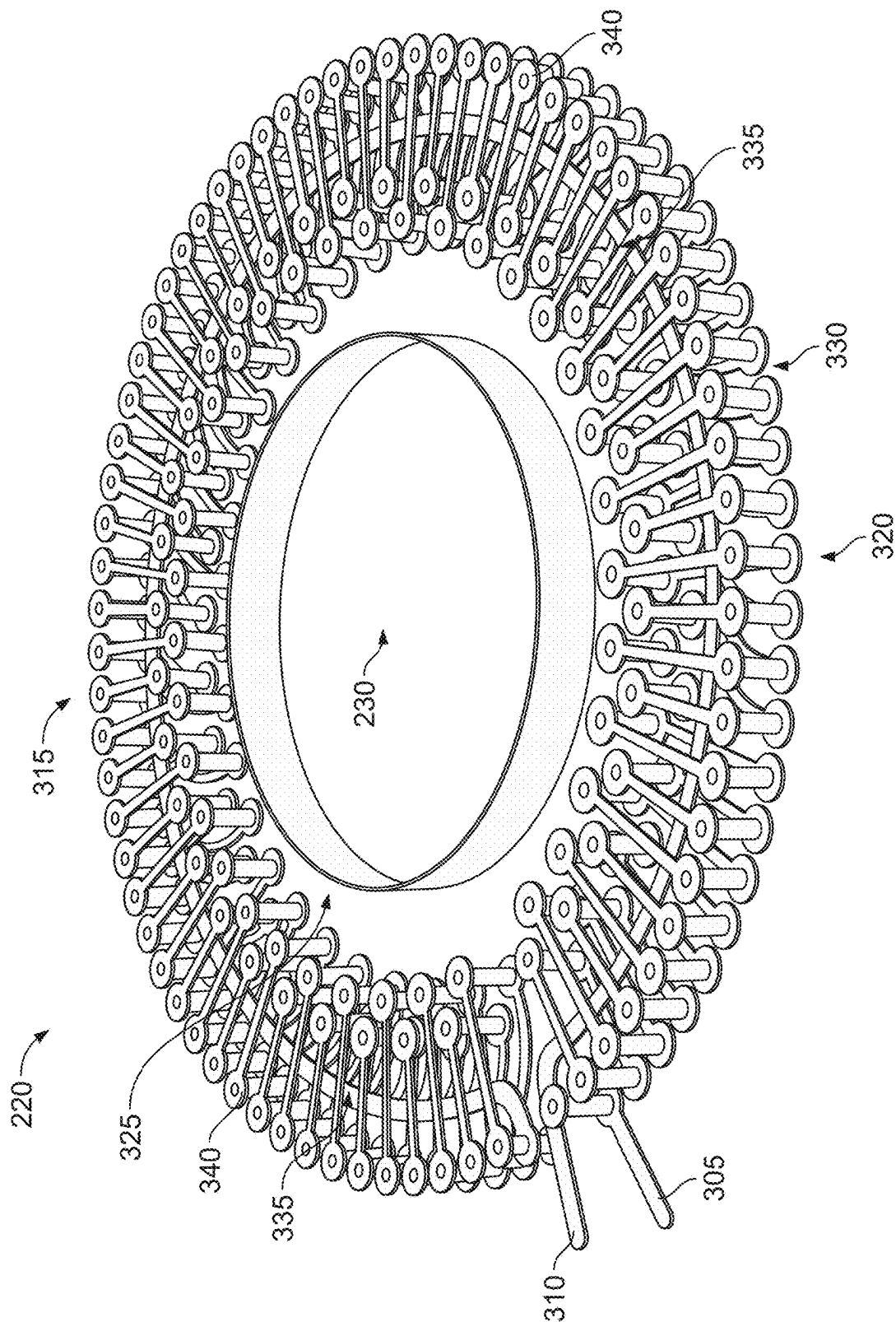


FIG. 3

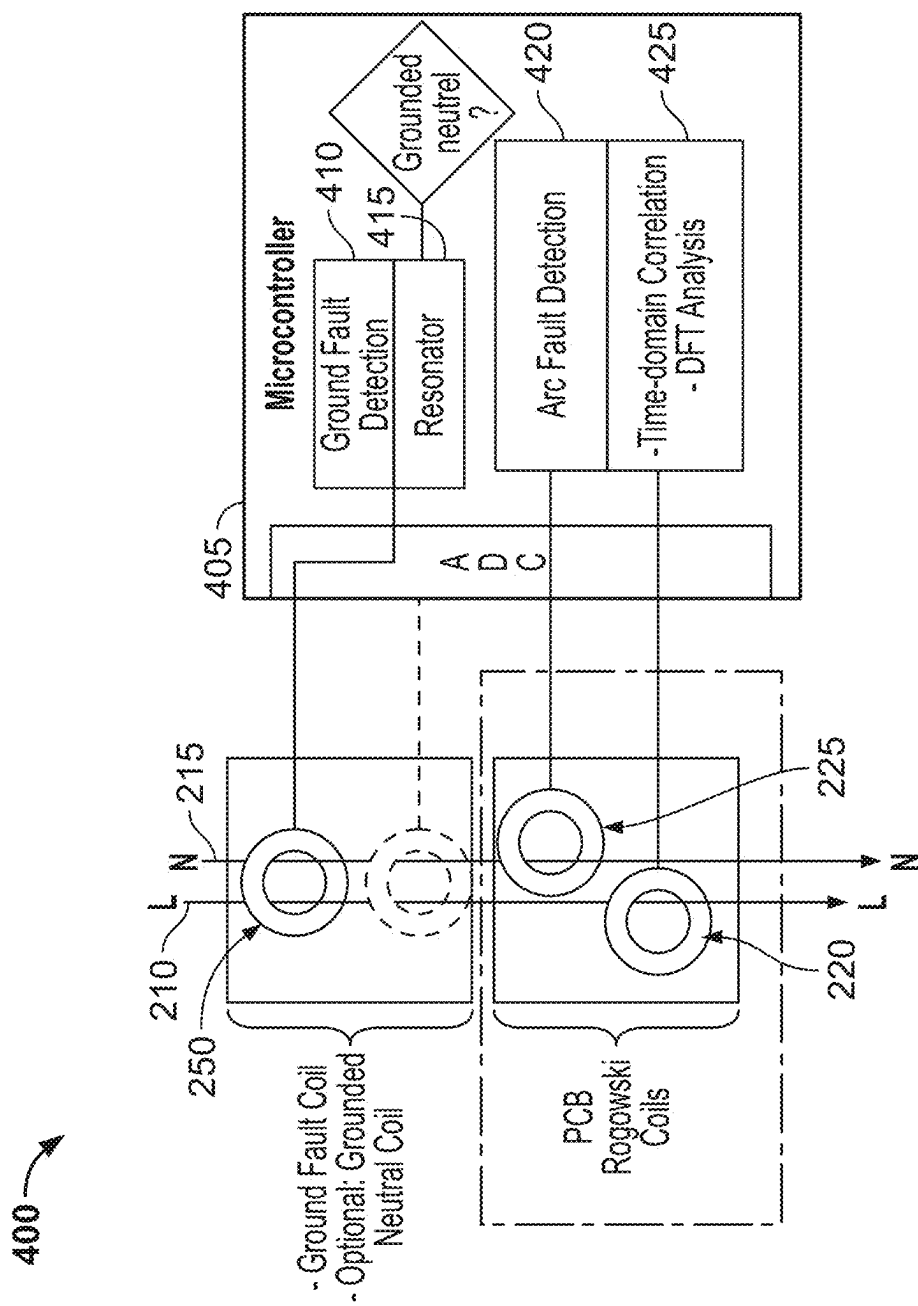


FIG. 4

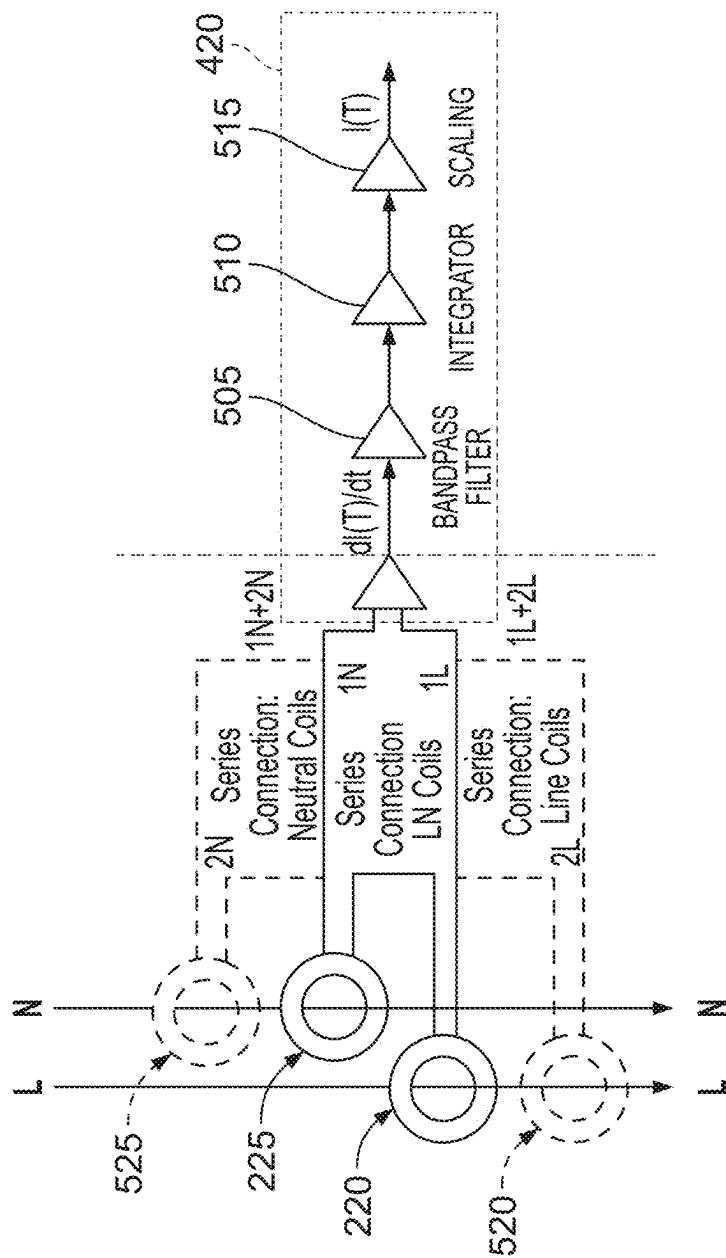


FIG. 5

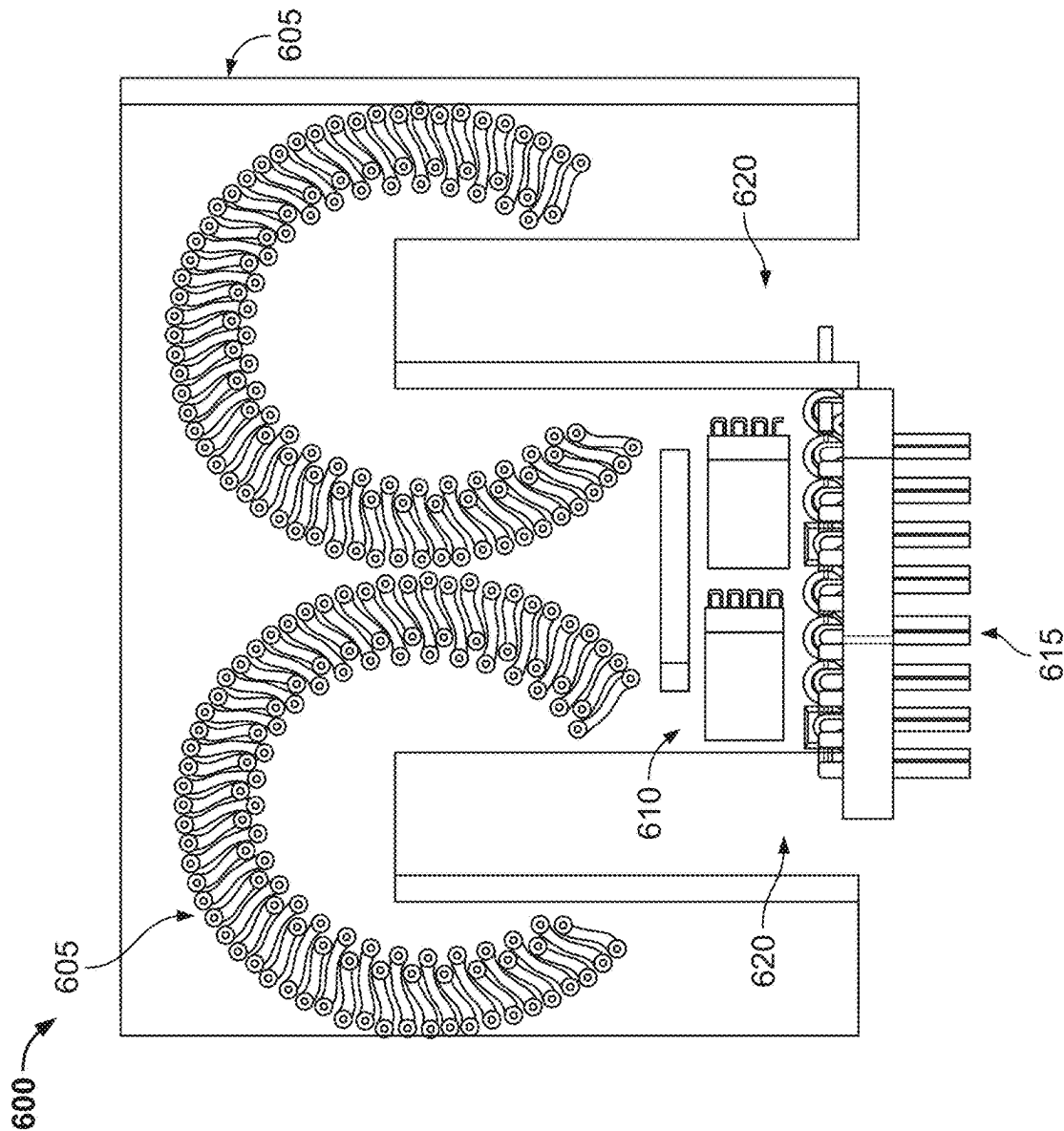


FIG. 6

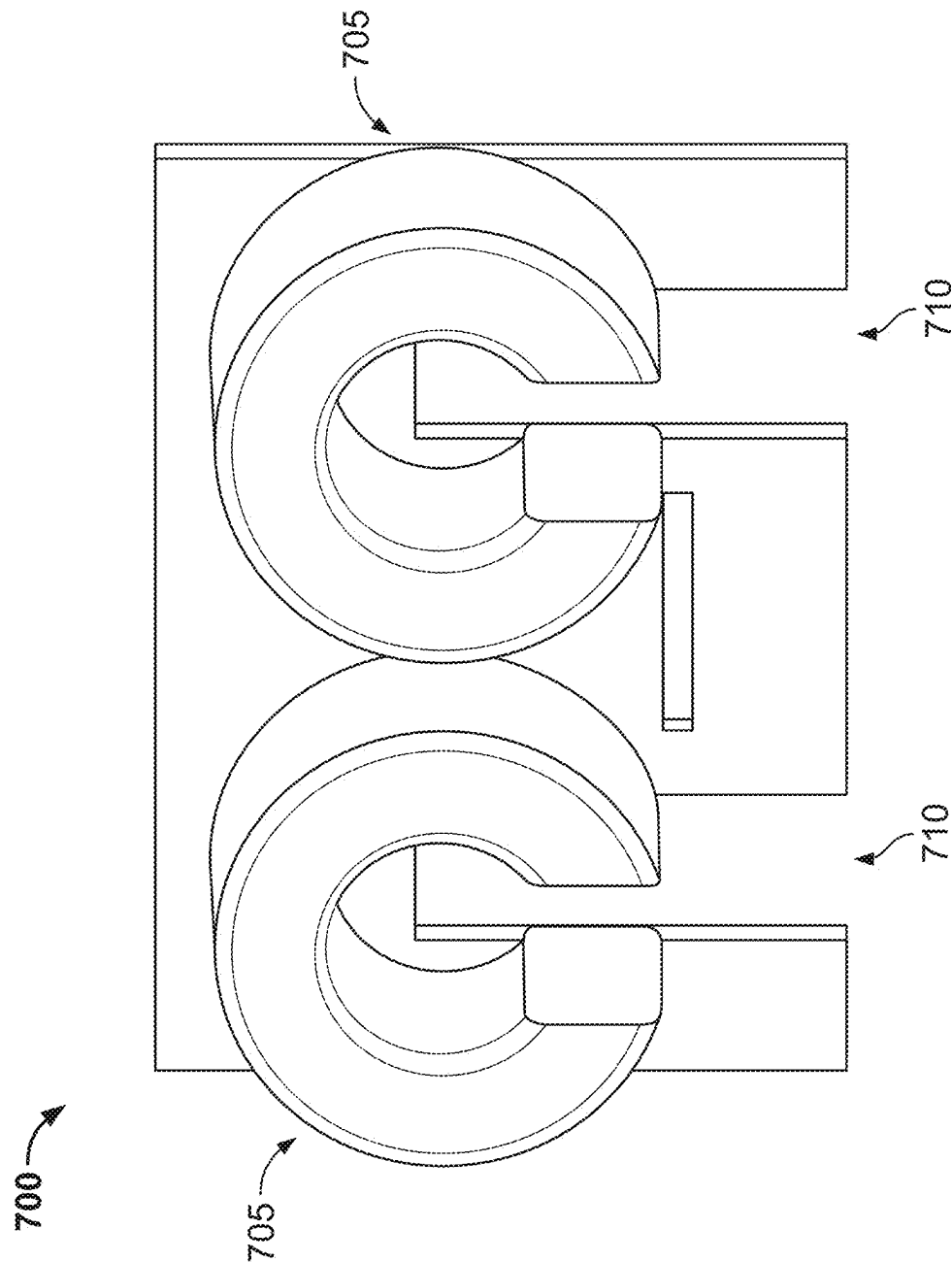


FIG. 7

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CIRCUIT INTERRUPTING DEVICE HAVING PRINTED CIRCUIT BOARD COILS

RELATED APPLICATIONS

This application claims priority to Ser. No. 17/985,758, filed on Nov. 11, 2022, which claims priority to U.S. patent application Ser. No. 16/520,588, filed Jul. 24, 2019, which claims priority to U.S. Provisional Patent Application No. 62/703,106, filed on Jul. 25, 2018, the entire contents of both of which are incorporated herein by reference.

FIELD

Embodiments relate to circuit interrupting devices, such as a ground fault circuit interrupter (GFCI) and/or an arc fault circuit interrupter (AFCI).

SUMMARY

Circuit interrupters are safety devices intended to protect a user from electric shock. GFCIs sense an imbalance in current flowing between hot and neutral conductors, and cut off power to the load, while AFCI sense an arc fault, and cut off power to the load. GFCI and/or AFCI may be implemented into electrical receptacles. In such an implementation, space within the electrical receptacle may be an issue.

Thus, one embodiment provides a circuit interrupter including a line conductor, a printed-circuit board, and a test circuit. The printed-circuit board coil is embedded on a printed circuit board. The printed circuit board includes a slot through the printed-circuit board coil, wherein the line conductor is received by the printed-circuit board coil. The test circuit is electrically connected to the printed-circuit board coil. The test circuit is configured to determine a condition of the line conductor based on a signal of the printed-circuit board coil.

Other aspects of the application will become apparent by consideration of the detailed description and accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective cutaway view of a circuit interrupting device according to some embodiments.

FIGS. 2A and 2B are perspective views of a core assembly of the circuit interrupting device of FIG. 1 according to some embodiments.

FIG. 3 is a perspective view of a coil of the circuit interrupting device of FIG. 1 according to some embodiments.

FIG. 4 is a block diagram of a control system of the circuit interrupting device of FIG. 1 according to some embodiments.

FIG. 5 is a block diagram of an arc fault detector of the control system of FIG. 4 according to some embodiments.

FIG. 6 is a perspective view of a printed-circuit board of a circuit interrupting device according to some embodiments.

FIG. 7 is a perspective view of a printed-circuit board of a circuit interrupting device according to some embodiments.

DETAILED DESCRIPTION

Before any embodiments of the application are explained in detail, it is to be understood that the application is not

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limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The application is capable of other embodiments and of being practiced or of being carried out in various ways.

FIG. 1 is a perspective cutaway view of a circuit interrupting device 100 according to some embodiments. The circuit interrupting device 100 includes a housing 105 having a front cover 110 and a rear cover 115. The housing 105 may be formed of plastic, or a similar material.

The front cover 110 may include a duplex outlet face 120 with a phase opening 125, a neutral opening 130, and a ground opening 135. The face 120 may further include an opening 140 accommodating a RESET button 145.

Although not illustrated, in some embodiments, the face 120 may include additional openings to accommodate additional buttons (for example, a TEST button), as well as additional openings to accommodate various indicators (for example, light-emitting diodes (LEDs), buzzers, etc.). The rear cover 115 is secured to the front cover 110 and may include one or more terminal screws 150. In some embodiments, the terminal screws 150 include a line terminal screw, a neutral terminal screw, and/or a ground terminal screw. Contained within the front and rear covers 110, 115 is a manifold 155. Manifold 155 provides support for a yoke/bridge assembly 165 configured to secure the device 100 to an electrical box.

FIGS. 2A and 2B illustrate perspective views of a core assembly 200 according to some embodiments. The core assembly 200 is configured to support a printed circuit board 205 that supports most of the working components of the device 100, including the control system 400 illustrated in FIG. 4. The core assembly 200 further supports a line conductor 210 and a neutral conductor 215. The line and neutral conductors 210, 215 are respectively electrically connected to the line terminal and neutral terminal, and are configured to supply electrical power to the device 100.

The core assembly 200 may further support a first coil 220 and a second coil 225. As illustrated, the first and second coils 220, 225 may respectively include first and second apertures 230, 235. In some embodiments, the first aperture 230 is configured to receive the line conductor 210, while the second aperture 235 is configured to receive the neutral conductor 215. In some embodiments, the first and second coils 220, 225 may respectively be embedded into first and second printed circuit boards 240, 245. In other embodiments, the first and second coils 220, 225 may be embedded into a single printed circuit board. In some embodiments, the first coil 220 and the second coil 225 are printed circuit board coils.

The core assembly 200 may additionally support a third coil 250 having a third aperture 255. In some embodiments, the third aperture 255 is configured to receive both the line conductor 210 and the neutral conductor 215.

FIG. 3 illustrates one embodiment of the first coil 220 with the printed circuit board removed for illustrative purposes. As illustrated, the first coil 220 may be a Rogowski coil having an input 305 and an output 310. As illustrated, the coil 220 further includes an upper portion 315, a lower portion 320, an inner portion 325, an outer portion 330, a plurality of helical conductors 335, and a plurality of nodes 340, connecting the input 305 to the output 310. As illustrated, the helical conductors 335, along with the nodes 340, form the coil 220. For example, the plurality of conductors 335 form a portion of the coil 220 between the inner portion 325 and the outer portion 330, while the plurality of nodes 340 form the coil 220 between the upper portion 315 and the lower portion 320.

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In some embodiments, the second coil **225** is also Rogowski coil, similar to coil **220**. Although not illustrated, in some embodiments the third coil **250** may also be a Rogowski coil embedded on a printed circuit board (for example a third printed circuit board or a single printed circuit board including the first, second, and third coils **220**, **225**, **250**). In some embodiments, coils **220**, **225**, and/or **250** are printed-circuit board coils that do not have a Rogowski coil configuration.

FIG. **4** is a block diagram of a control system, or testing circuit, **400** of the device **100** according to some embodiments. The control system **400** includes a controller, or microcontroller, **405** electrically connected to the first coil **220**, the second coil **225**, and the third coil **250**. The controller **405** is configured to detect one or more fault conditions, and place the device **100** into a tripped state when the one or more fault conditions are detected. In some embodiments, the controller **405** is a well-known integrated circuit device having an electronic processor and a memory, such as but not limited to a **4145** device.

The controller **405** may include a ground fault detection unit **410**, a resonator **415**, an arc fault detection unit **420**, and a time-domain correlator and analyzer **425**. In some embodiments, the ground fault detection unit **410**, the resonator **415**, the arc fault detection unit **420**, and/or the time-domain correlator and analyzer **425** are implemented in whole or in part in software. In some embodiments, there is no separate module, but rather the ground fault detection unit **410**, the resonator **415**, the arc fault detection unit **420**, and/or the time-domain correlator and analyzer **425** are implemented using software stored in the memory of the controller **405** and executed by the processor of the controller **405**.

The ground fault detection unit **410** is configured to analyze electric signals from the third coil **250**. The ground fault detection unit **410** is configured to detect a ground fault (for example, a real ground fault, a simulated ground fault, a self-test ground fault, and/or a real or simulated grounded neutral fault based on the electric signals from the third coil **250**). The resonator **415** is configured to analyze a frequency of the power supplied to the device **100**.

The arc fault detection unit **420** is configured to analyze electric signal from the first coil **220** or from the first coil **220** and second coil **225**. The arc fault detection unit **420** is configured to detect an arc fault (for example, a real arc fault, a simulated arc fault, and/or a self-test arc fault) based on the electric signals from the first coil **220** or from the first coil **220** and second coil **225**. The time-domain correlator and analyzer **425** is configured to perform a time-domain transformation and/or analysis on the electric signals from the first coil **220** or from the first coil **220** and second coil **225**. The transformed electric signals are then analyzed by the arc fault detection unit **420** to detect an arc fault. In some embodiments a discrete Fourier transform (DFT) is performed on the electric signal and then analyzed to further determine an arc fault.

FIG. **5** is a block diagram of the arc fault detection unit **420** according to some embodiments. In such an embodiment, the arc fault detection unit **420** includes a bandpass filter **505**, an integrator **510**, and a gain stage, or scaling module, **515**. The electric signals from the first coil **220** or from the first coil **220** and second coil **225** are filtered by the bandpass filter **505** and then integrated by integrator **510** in order to determine a voltage of the electric signal(s). In some embodiments, the voltage is proportional to a current flowing through the first coil **220** and/or the second coil **225**. In some embodiments, the bandpass filter **505** is a 3-dB pass-band filter between 1-Hz and 8-kHz, which attenuates

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unnecessary low and high frequency content that might otherwise saturate the integrator **510**. Once integrated, the gain stage **515** scales the signal to a full-scale input voltage of an analog-to-digital (A/D) converter, which will sample the signal for subsequent digital post-processing. For example, a 30-Arms line-current may be scaled to a full-scale voltage of approximately 3.0 Vdc by the A/D converter. In some embodiments, the A/D converter is embedded within the controller **405**.

As illustrated in FIG. **5**, in some embodiments, the interrupter **100** may further include coils **520**, **525**. Coil **520** may be electrically connected to coil **220** in a series-type configuration, while coil **525** may be electrically connected to coil **225** in a series-type configuration. Coils **520**, **220** and coils **525**, **225**, when respectively electrically connected in a series-type configuration, may produce respective measured signals that are multiplied by a number of coils connected in the series-type configuration. Such an embodiment may allow for a greater measured signal.

FIG. **6** illustrates a printed-circuit board **600** according to some embodiments. Printed-circuit board **600** may be part of, or included in, circuit interrupting device **100**. The printed-circuit board **600** may include one or more printed-circuit board coils **605**, one or more electronic components **610**, and one or more electrical pins **615**. Printed-circuit board coils **605** may be substantially similar to coils **220**, **225**, and/or **250**. The one or more electrical components **610** may include one or more components discussed above with respect to FIGS. **4** and **5**. For example, the one or more electrical components **610** may be, or may include, a programmable microcontroller. The one or more electrical pins **615** may be configured to electrically and/or communicatively connect the printed-circuit board **600** to other components of the circuit interrupting device **100**.

In the illustrated embodiment, printed-circuit board **600** further includes one or more slots, or apertures, **620**. The slots **620** may be configured to receive the line conductor **210** and/or neutral conductor **215**.

FIG. **7** illustrates a printed-circuit board **700** according to other embodiments. Printed-circuit board **700** may be part of, or included in, circuit interrupting device **100**. The printed-circuit board **700** may include one or more coils **705** and one or more slots, or apertures, **710**. In the illustrated embodiment, the one or more coils **705** are wire-wound coils. The slots **710** may be configured to receive the line conductor **210** and/or neutral conductor **215**.

In operation, the coils (for example, coils **220**, **225**, **250**, **605**, and/or **705**) may be used to sense and/or monitor a current. An arc condition may then be determined based on the current. In some embodiment, an arc condition may be determined by determining if a correlation condition, a volatility condition, and/or an impulse condition exists. Additionally, in some embodiments, an in-rush condition may be detected via the coils.

Thus, the application provides, among other things, a circuit interrupting device having a printed circuit board coil. Various features and advantages of the application are set forth in the following claims. For example, one advantage of the application includes an increase in within an electrical receptacle due to the reduced footprint of using one or more printed circuit board coils.

What is claimed is:

1. A testing system comprising:

- a printed-circuit board coil embedded on a printed circuit board; and
- a test circuit electrically connected to the printed-circuit board coil, the test circuit configured to:

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receive a signal from the printed-circuit board coil, analyze the signal from the printed-circuit board coil, and determine at least one selected from a group consisting of a:

a volatility condition, an impulse condition, and an in-rush condition based on the signal from the printed circuit board coil.

2. The testing system of claim 1, wherein the analyzing of the signal from the printed-circuit board includes a time-domain transformation.

3. The testing system of claim 1, wherein the analyzing of the signal from the printed-circuit board includes performing a Fourier transform on the signal.

4. The testing system of claim 1, wherein the analyzing of the signal from the printed-circuit board includes analyzing at least one selected from a group consisting of a voltage, a current, and a frequency of the signal.

5. The testing system of claim 1, wherein the printed circuit board includes a slot through the printed-circuit board coil.

6. The testing system of claim 5, wherein a conductor is received within the slot.

7. The testing system of claim 1, wherein the printed-circuit board coil is a Rogowski coil.

8. The testing system of claim 1, wherein the analyzing of the signal includes filtering the signal.

9. The testing system of claim 8, wherein the filtering of the signal includes using a bandpass filter.

10. The testing system of claim 9, wherein the bandpass filter is a 3-dB pass-band filter between 1-Hz and 8-kHz.

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11. A method of testing a signal comprising: receiving, at a test circuit, the signal from a printed-circuit board coil embedded on a printed circuit board; analyzing, via the test circuit, the signal from the printed-circuit board coil; and

determining, via the test circuit, at least one selected from a group consisting of a volatility condition, an impulse condition, and an in-rush condition based on the signal from the printed circuit board coil.

12. The method of claim 11, wherein the analyzing of the signal from the printed-circuit board includes a time-domain transformation.

13. The method of claim 11, wherein the analyzing of the signal from the printed-circuit board includes performing a Fourier transform on the signal.

14. The method of claim 11, wherein the analyzing of the signal from the printed-circuit board includes analyzing at least one selected from a group consisting of a voltage, a current, and a frequency of the signal.

15. The method of claim 11, wherein the printed circuit board includes a slot through the printed-circuit board coil.

16. The method of claim 15, wherein a conductor is received within the slot.

17. The method of claim 11, wherein the printed-circuit board coil is a Rogowski coil.

18. The method of claim 11, wherein the analyzing of the signal includes filtering the signal.

19. The method of claim 18, wherein the filtering of the signal includes using a bandpass filter.

20. The method of claim 19, wherein the bandpass filter is a 3-dB pass-band filter between 1-Hz and 8-kHz.

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